Comprehensive Supervisor Review Report: Pre-Defense 1 Milestone

Project Title: Development of a Compatibility-Based Peer Learning Platform **Student:** Жумиров Бекзат Ерболатулы (Zhumirov Bekzat) | Student ID: `241390` **Academic Email:** `241390@astanait.edu.kz` **Group / Programme:** SE-2416, Software Engineering **Academic Supervisor:** Aigerim Kydyrbekova (Айгерим Кыдырбекова) **Academic Milestone:** Pre-Defense 1 (Working Review Checkpoint: 20 October 2026; Repository Milestone: 24 October 2026) **Remote Git Repository:** https://github.com/belozov/peer-learning-platform **Active Task Branch:** `PLP-16-research-literature-and-evaluation` **YouTrack Cloud Instance:** https://bekzat-diploma-2027.youtrack.cloud

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1. Executive Summary & Delivery Scope

This comprehensive report compiles the complete body of engineering analysis, requirements specification, academic research, architectural design, and prototype implementation developed for the **Pre-Defense 1** milestone of the individual diploma project.

The central problem addressed by this research is **coordination and compatibility friction in self-directed peer learning**. While modern students frequently seek collaborative study partners for technical disciplines (such as programming, calculus, and language practice), existing tools (generic messaging apps, Discord servers, and Q&A boards) force learners to endure high coordination overhead, lack structured schedule alignment, provide no compatibility filtering, and fail to track study commitments.

The proposed platform provides an integrated, non-commercial solution built around:

- ♦ **Community Channel Gating**: Scoped subject communities and topic channels (FR-03).
- ♦ **Explainable Multi-Attribute Compatibility**: Deterministic scoring (S = 0.40*T* + *0.35*G + 0.25*L) presenting human-readable justifications (`Score: 80.0 / 100`) without black-box machine learning.
- ♦ **Accounted Synchronous Study Sessions**: Time-bounded study sessions backed by a confirmed 180-minute daily allowance and presence tracking (FR-10, FR-11).
- ♦ **Version-Safe Collaborative Outcomes**: Post-session co-documentation of covered topics requiring mutual confirmation (FR-12).

Delivery Boundaries: Prototype vs. Working-V1

To maintain strict truth-in-reporting, the project separates its delivery roadmap:

- **Milestone Demonstration Prototype (Batch 4 — Delivered & Verified)**: A complete, runnable client-side application built in standard HTML5, CSS3, and JavaScript (universal UMD module). It features an in-memory transactional store, deterministic demo clock, reconciler for canonical accounting fixtures (A-01..A-04, A-06, S-01, M-01), channel gating, and outsider privacy restrictions. It is verified by **13/13 passing automated test suites in Node.js executing `app.js` directly**.
- **Proposed Working-V1 Production Architecture (Specified)**: A modular monolithic backend in Node.js/Express (TypeScript), PostgreSQL 16 relational database with row-level locking (`SELECT FOR UPDATE`), Native WebSockets (`ws`), and React 18/19 SPA.

Epistemological Integrity & Pending Evidence

- **Pending Adult Fieldwork (Component 2)**: Semi-structured interviews with 3–5 adult learners (>= 18) remain pending participant scheduling. Protocols, ethics safeguards, and recording templates are fully prepared in `docs/user-research.md`. No interview data has been fabricated.
- **YouTrack Cloud Synchronization**: **Verified & Synchronized**. Project `PLP` (https://bekzat-diploma-2027.youtrack.cloud/projects/PLP) hosts all 22 milestone tasks (`PLP-1` through `PLP-22`) with full requirements-traceable acceptance criteria.

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2. Research Evidence & Literature Review

2.1 Bibliographic Synthesis

The literature review surveyed **14 verified scholarly publications**, satisfying and exceeding university criteria (>= 12 sources, >= 8 scientific):

- **11 Peer-Reviewed Scientific Journal Articles** (IEEE TLT, IEEE TE, ACM TOCHI, ACM TACCESS, PACM HCI/CSCW, Computers & Education, ijCSCL, Review of Educational Research, Educational Psychology, Topics in Cognitive Science, and PLOS ONE).
- **2 Scholarly Books / Book Chapters** (Elsevier Science, Routledge).
- **1 Foundational Academic Monograph** (Harvard University Press).
- **Recent Research (2021–2024)**: 7 publications (50.0% of citations).

All entries were audited against CrossRef/publisher DOIs. Previously identified fabricated citations (Computers in Human Behavior 107, Computers & Education 158) were permanently expunged, and 3 unverified entries (Zheng, Borchers, Rientes) were quarantined.

2.2 Thematic Matrix & Engineering Connections

Thematic Area	Key Citations	Core Findings	Engineering Implementation
Symmetrical Collaboration	Topping (2005), Dillenbourg (1999), Boud et al. (2001/2014)	Symmetrical problem-solving and absence of evaluative anxiety drive cognitive gains. Commercial hierarchies erode mutual accountability.	**DEC-001 & Scope §4**: All users hold uniform `Learner` role. Author and Respondent have equal agency; no paid tutoring.
Interactive Discourse & Body Doubling	Chi (2009), Roscoe & Chi (2007), Eagle et al. (2024)	Interactive dialogue triggers constructive cognitive updates. Virtual co-presence ("body doubling") acts as an external commitment device.	**FR-10, FR-11, FR-17**: Real-time synchronous text chat; deterministic 180-min daily allowance; 60s reconnection buffer.
Algorithmic Compatibility	Vygotsky (1978), Revelo-Sánchez et al. (2021), Vallès-Català & Palau (2023), Saksono et al. (2023)	Multi-criteria pairing outperforms random search. Black-box ML harms learner trust and agency.	**FR-07**: Multi-attribute formula S = 0.40*T* + *0.35*G + 0.25*L with transparent breakdown and chronological fallback (C-04).
Reciprocal Feedback & Outcomes	Kulkarni et al. (2013), Misiejuk & Wasson (2021), Tan et al. (2023), Avila et al. (2022)	Closing the feedback loop via structured rubrics and backward evaluation mitigates rating bias.	**FR-12, FR-13**: Version-safe outcome co-confirmation; 300s review eligibility gate (S-01) preventing zero-second review spam.

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3. Existing Systems Comparative Analysis

Five platforms were evaluated across ten uniform criteria (C1–C10, established in `docs/project-scope.md` §5) using official documentation and October 2026 recorded observations:

```
COMMUNICATION DYNAMICS
Asynchronous Synchronous
+----------------------+----------------------+
General | Discord | Focusmate |
Productivity | (Unstructured | (Time-bounded body |
| channel chat) | doubling, no topics)|
COORDINATION +----------------------+----------------------+
Academic & | Brainly | **PROPOSED PLATFORM**|
Pedagogical | (Asynchronous Q&A, | (Structured listings, |
| no live study) | compatibility score, |
| | bounded study time) |
+----------------------+----------------------+
```

Comparative Summary Matrix

Criterion	Focusmate	Tandem	Brainly	Discord (Study Servers)	Studypool	Proposed Platform (Working-V1)
C1: Community Structure	No	Partial	Yes	Yes	Yes	**Yes (FR-03)**
C2: Structured Listings	Partial	No	No	Partial	Yes	**Yes (FR-06)**
C3: Multi-Attribute Matching	No	Partial	No	No	No	**Yes (0.40*T* + *0.35*G + 0.25*L)**
C4: Explainability of Matches	No	No	No	No	No	**Yes (Score: X/100, FR-07)**
C5: Knowledge Level Handling	No	Partial	No	No	Partial	**Yes (FR-02/06)**
C6: Availability Gating	Yes	No	No	No	No	**Yes (Half-Open Intervals)**

Criterion	Focusmate	Tandem	Brainly	Discord (Study Servers)	Studypool	Proposed Platform (Working-V1)
C7: Accounted Synchronous Time	Yes	No	No	Partial	No	**Yes (10,800s Daily Cap)**
C8: Mutual Outcome Co-Approval	Partial	No	No	No	No	**Yes (Version-Safe, FR-12)**
C9: Default Privacy & Safety	Partial	Partial	No	Partial	No	**Yes (Default-Private, FR-15)**
C10: Non-Commercial Reciprocity	Partial	Partial	Partial	Yes	No	**Yes (DEC-001)**

Identified Practical Opportunity

Existing tools force learners into an unsatisfactory compromise: generic chat tools (Discord) provide rich communities but force learners to endure tedious, manual coordination friction; specialized tools (Focusmate) enforce time accounting but are subject-blind. The proposed platform bridges this gap by coupling **structured subject listings** and **explainable compatibility ranking** with **bounded synchronous study sessions**.

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4. Requirements & Lifecycle Specification

4.1 Requirements Catalog Summary

The specification catalog (`docs/requirements-spec.md`) details 19 functional and 7 non-functional requirements with two-tier acceptance criteria:

- **FR-01..FR-04**: Account profiles, learning preferences, community/channel hierarchy, and membership gating.
- **FR-05..FR-09**: Structured study listings, multi-attribute matching, response submission/withdrawal reuse, and atomic single-partner acceptance.
- **FR-10..FR-13**: Real-time synchronous chat, 180-minute daily allowance accounting, version-safe outcome confirmation, and discrete peer reviews.
- **FR-14..FR-17**: Profile history, default-private partner visibility, active session blocking, and presence tracking with 60s grace buffer.
- **NFR-01..NFR-07**: Password hash privacy invariant, transactional integrity (Shared Lock Protocol), response latency (<= 200ms), mobile responsive layout (360px–1440px), and adult ethical safeguards (>= 18).

4.2 State Machine & Time Accounting (Invariant #13 & #14)

Sessions transition strictly through three states: `scheduled` → `active` → `ended`.

- **Fixed End Boundary**: `scheduled_end = scheduled_start + duration`. Late starts do not extend the end boundary.
- **Exact-Midnight Policy (Fixture A-01)**: Sessions running across midnight are split into daily ledgers. If Day 1 allowance exhausts at 00:00:00 UTC, the session terminates immediately with `allowance_exhaustion`; Day 2 reset never revives an ended session.
- **Terminal Reason Priority**: When multiple boundaries coincide, deterministic precedence resolves the reason: (1) `blocked`, (2) `allowance_exhaustion`, (3) `scheduled_end`, (4) `connection_interruption`, (5) `manual_exit`.

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5. Architectural Blueprints & Data Model

5.1 Architecture Overview

The system is architectured as a **Modular Monolith** backing a Single Page Application:

- **Client Layer**: React 18/19 SPA with TypeScript and Vite.
- **API & Gateway Layer**: Node.js and Express 5 co-hosting REST endpoints and native WebSocket server (`ws`).
- **Domain Modules**: Account & Auth, Community & Channels, Discovery & Matching, Session Lifecycle & Accounting, and Outcomes & Reviews.
- **Persistence Layer**: PostgreSQL 16 relational database with GIN array indexing (`TEXT[]`) and row-level locking.

5.2 Architectural Diagrams (Mermaid Source & SVG Exports)

- **System Context Diagram** (`diagrams/exports/system-context.svg`): Delineates system boundaries, external browser clients, and explicit exclusions (zero external payment or third-party video dependencies).
- **Use Case Diagram** (`diagrams/exports/use-case.svg`): Structured with UML use-case semantics: channel membership modeled as a strict precondition (`«precondition»`) before listing/discovery; session start decoupled from acceptance; review decoupled from start; added administrator channel creation (`UC-20`).
- **Component Diagram** (`diagrams/exports/component.svg`): Labelled *Proposed Working-V1 Architecture*, detailing internal module interfaces, event bus, and database repositories.

5.3 Shared Lock Protocol (Invariant #3 & #4)

To prevent concurrent booking race conditions (NFR-02), all booking paths acquire locks in strict global ascending order: users (UUID ASC) -> study_listings (ID ASC) -> listing_responses (ID ASC) -> study_sessions (ID ASC) -> session_outcomes (ID ASC) Acquiring both participant user locks upfront prevents concurrent overlapping sessions across competing listings.

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6. Demonstration Prototype Evidence (Batch 4)

The interactive prototype (`prototype/`) demonstrates the complete end-to-end user journey in a self-contained, client-side web application.

6.1 Automated Test Suite Verification (`prototype/test-prototype.js`)

The test harness evaluates the actual `prototype/app.js` application controller directly via its universal UMD export:

```
> peer-learning-platform-prototype@1.0.0 test
> node test-prototype.js

================================================================
RUNNING PROTOTYPE AUTOMATED VERIFICATION SUITE
Application Controller: prototype/app.js (UMD module)
Student: Zhumirov Bekzat (SE-2416) | Supervisor: Aigerim Kydyrbekova
================================================================

[PASS] Test 1: Compatibility Formula Exactness & Format (Fixture M-01: 60.8 / 100)
[PASS] Test 2: Invariant #10: Missing Preferences Suppress Scoring Without Invented Values
[PASS] Test 3: Priority 1: Channel Membership Gating & Join/Leave Interactions
[PASS] Test 4: Priority 1: Strict Listing Validation (Future Start, Positive Duration, Language, Author Level)
[PASS] Test 5: Invariant #8 & Fixture R-01: Response Row Reuse on Withdrawal (UNIQUE(listing_id, respondent_id))
[PASS] Test 6: Priority 1 & Invariant #5: Acceptance Creates Scheduled Session (Not Immediately Active)
[PASS] Test 7: Priority 2: Outsider Persona Cannot Interfere with Private Session
[PASS] Test 8: Canonical Fixture A-01: Exact-Midnight Exhaustion (900s Day 1, None on Day 2)
[PASS] Test 9: Canonical Fixtures A-02 & A-03: Across-Midnight Split (2,700s) & Midnight Scheduled End
[PASS] Test 10: Canonical Fixture A-04: End / Exhaustion Tie Priority Rule (allowance_exhaustion wins)
[PASS] Test 11: Canonical Fixtures A-06 & S-01: 31-Second Session & Review Eligibility Gate
[PASS] Test 12: Invariant #17 & Fixtures O-01..O-03: Version-Safe Collaborative Outcomes & 409 Conflict
[PASS] Test 13: Invariants #18 & #19: Exact 5 Terminal Reasons & Blocking Guard

================================================================
TEST SUITE RESULTS: 13/13 TESTS PASSED (100%)
Prototype Test Suite Completed: Verified client-side interactions, canonical fixtures,
and state machine guards (Production PostgreSQL concurrency & WebSockets remain Working-V1 scope).
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```

6.2 The 9-Step Demonstrated Walkthrough

- **Channel Membership Gating**: Visiting *Algorithms* channel locks *Find a Partner* until clicking **Join Channel**.
- **Score Formatting**: Bob's listing displays `Compatibility score: 80.0 / 100` (points out of 100; percentage symbol `%` is completely absent).
- **Missing Preference Fallback**: Switching to Eve (lacks topic tags) automatically switches feed sorting to chronological order (`created_at DESC, id ASC`) and updates the indicator to *Chronological (Fallback due to missing preferences)*.
- **Strict Listing Validation**: Creating a listing enforces future start relative to demo clock, end after start, positive duration (15–180m), language, and explicit author level.
- **Response & Scheduled Acceptance**: Alice accepts Bob's response. The session is created in `scheduled` state with accounted duration 0s (does not immediately become active).
- **Demo Clock Advance & Check-In**: Alice advances the demo clock to `2026-10-25 14:00 UTC` and clicks **Check-in & Start**. Session transitions to `active`.

- ♦ **Outsider Access Restriction**: Switching to Eve while Alice–Bob session is active renders a prominent access restricted notice, hiding chat messages, outcome editing, and reviews.
- ♦ **Deterministic Reconciler**: Dedicated control buttons execute canonical fixtures A-01, A-02, A-03, A-04, A-06, and S-01, reconciling daily HUD, session duration, and ledger entries simultaneously.
- ♦ **Version-Safe Outcomes**: Alice edits draft to v2 (invalidating Bob's confirmation). Bob clicking stale v1 confirmation receives `409 OUTCOME_VERSION_CONFLICT`. Bob approving v2 confirms outcome.

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7. Standalone Empirical Evaluation Protocol

The evaluation plan (`docs/evaluation-plan.md`) establishes a formal research protocol addressing three research questions:

- **RQ1 (Discovery Efficiency & Selection Validity)**: Evaluates whether compatibility ranking increases valid partner selections and reduces time-to-selection (T_select) compared to chronological browsing.
- **RQ2 (Algorithmic Transparency & Comprehension)**: Measures whether participants accurately interpret score breakdowns and chronological fallback on an objective 3-item test (H2 >= 80%).
- **RQ3 (System Usability & Workload)**: Assesses overall usability across the 9-step journey using the standardized System Usability Scale (H3 > 68.0).

Methodological Controls (Invariant #11)

- **Within-Subjects Design**: Counterbalanced ($A \to B$ vs. $B \to A$) across standardized 20-listing pools.
- **Strictly Identical Filtering**: Both conditions apply **identical hard eligibility filters** (channel membership, date interval overlap, language, absence of blocks); only the ranking permutation differs.
- **Proposed Sample Size**: N = 10-20 adult volunteers (>= 18). Protocol is planned; zero pre-study findings are claimed.

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8. Project Governance, Backlog & Outstanding Work

8.1 11-Component Compliance Status

№	Component	Primary Deliverable	Current Status
1	Project Proposal	`docs/project-proposal.md`	**Drafted**
2	Evidence of Relevance	`docs/user-research.md`	**Pending Adult Fieldwork**
3	Literature Review	`docs/literature-review.md`	**Completed (14 Sources)**
4	Existing Systems Comparison	`docs/existing-systems.md`	**Completed (5 Systems / C1–C10)**
5	Technology Comparison	`docs/technology-comparison.md`	**Completed (Revised)**
6	Requirements Specification	`docs/requirements-spec.md`	**Drafted (Revised)**
7	Architecture & Diagrams	`docs/architecture.md`, `diagrams/`	**Completed (Draft Exports)**
8	UX / Prototype	`prototype/`	**Delivered (Simulation Prototype)**
9	Verification & Evaluation Plan	`docs/evaluation-plan.md`	**Completed (Protocol)**
10	Project Management & Risks	`docs/youtrack-tasks.md`, `README.md`	**Completed (Synchronized & Tagged)**
11	Supervisor Feedback	`docs/supervisor-log.md`	**Pending Supervisor Review**

8.2 Open Architectural Decisions Awaiting Supervisor Guidance

- ♦ **Approval of Proposed Design Rules**: Multi-attribute weights (0.40*T* + *0.35*G + 0.25*L), UTC daily reset (00:00 UTC), 60s disconnection grace timer, 300s review threshold, and mutual outcome confirmation requirement.
- ♦ **Accounted Time Framing**: Explicit framing that session time measures accounted participation duration, not verified cognitive attention or grade improvement.
- ♦ **Adult Safeguard Protocol**: Strict restriction of preliminary fieldwork to adult participants (>= 18) pending institutional ethics clearance for minor recruitment.

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9. Review Package Verification & Launch Instructions

9.1 Launching Prototype Locally

```
# Option A: From repository root
python -m http.server 3000 --directory prototype
node prototype/test-prototype.js

# Option B: From prototype/ directory
cd prototype
python -m http.server 3000 --directory .
npm test
```

Access in browser: `http://localhost:3000`

9.2 Project Tracking, Remote Git & Review Archive

- **YouTrack Cloud Project**: https://bekzat-diploma-2027.youtrack.cloud/projects/PLP (Project Key: `PLP`, 22 milestone tasks `PLP-1`..`PLP-22` synchronized).
- **Remote Git Branch**: `PLP-16-research-literature-and-evaluation` on `https://github.com/belozov/peer-learning-platform`.
- **GitHub Pull Request**: https://github.com/belozov/peer-learning-platform/pull/2 (PR #2, open for supervisor review).
- **Official Submission Tag**: `prob-period-1`.
- **Required University Submission URL**: https://github.com/belozov/peer-learning-platform/tree/prob-period-1.
- **Review ZIP Archive**: `C:\Users\QWERTY\Documents\peer-learning-platform-batch-6-review.zip` (self-contained review package without `.git/` or `node_modules/`).